Structured notes: Module I: crystal structure and X-ray diffraction

Puri and Babbar, *Solid State Physics and Electronics*, chapters 1 and 2 (printed pages 1–39 and 46–58; section 2.3 is outside the module). Figures are redrawn for these notes, not copied from the book; the book figure number is in each caption. Problems are not included.

Section 1: crystal lattice, unit cell and basis (Puri and Babbar 1.1–1.4)

1.1–1.4 Crystal lattice, translation vectors, unit cell and basis (structured notes)

Source: book-sources/bs-1-crystal-structure.html, Puri and Babbar pp. 1–5. Equations 1.1 and 1.2 are cited by their book numbers. Figures are redrawn; the book figure number is in each caption.

Before you start

- Lattice: an infinite set of points with identical surroundings (book §1.2).
- Translation operation: $\mathbf{T}$ with whole-number coefficients on $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$.
- Triple product and cell volume (Plus Two recall below).

> **PLUS TWO RECALL: AREA AND VOLUME FROM VECTOR PRODUCTS**
>
> **Recall.** The cross product $\mathbf{a} \times \mathbf{b}$ has magnitude $ab\sin\theta$, the area of the parallelogram formed by $\mathbf{a}$ and $\mathbf{b}$.
>
> **Re-derive.**
>
> 1. Parallelogram with sides a, b and angle θ: base a, height $b\sin\theta$, area $ab\sin\theta = |\mathbf{a} \times \mathbf{b}|$.
> 2. Parallelepiped with third edge $\mathbf{c}$: volume = base area × height, height $= |\mathbf{c} \cdot \hat{\mathbf{n}}|$ with $\hat{\mathbf{n}} = (\mathbf{a} \times \mathbf{b})/|\mathbf{a} \times \mathbf{b}|$.
> 3. Multiply: $V = |\mathbf{a} \times \mathbf{b}|\,|\mathbf{c} \cdot \hat{\mathbf{n}}| = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$.
>
> **Saved.** $V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|$ (book §1.3, not numbered).

1.1 Definition of a lattice

A point lattice (space lattice) is an infinite array of imaginary points in three-dimensional space in which every point has identical surroundings. The lattice has the same appearance seen from any of its points. This holds only if the points form a repeating pattern unit, moved by a translation operation.

1.2 Translation vectors

The translation operation is

$$\mathbf{T} = n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c} \tag{1.1}$$

where n_1, n_2, n_3 are arbitrary integers and $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ are the fundamental translation vectors. The relation between two lattice points is

$$\begin{aligned}\mathbf{r}' &= \mathbf{r} + \mathbf{T} \\ &= \mathbf{r} + n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c}\end{aligned} \tag{1.2}$$

A perfect lattice satisfies Eq. (1.2) for every integer choice. An imperfect lattice does not admit such vectors.

> **RESULT 1: PRIMITIVE AND NON-PRIMITIVE VECTORS**
>
> Vectors whose translation operation has integral coefficients are *primitive*. Vectors that need non-integral coefficients, for example $\mathbf{R}' = \mathbf{R} + \frac{1}{2}\mathbf{a}_2 + \frac{1}{2}\mathbf{b}_2$, are *non-primitive*. The book prefers an orthogonal, shortest set (Fig. 1.1).

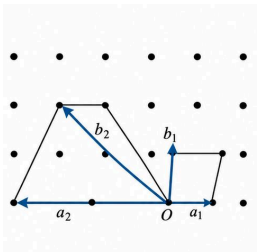


Fig. 1.1 (book): primitive ($\mathbf{a}_1$, $\mathbf{b}_1$) and non-primitive ($\mathbf{a}_2$, $\mathbf{b}_2$) translation vectors in a two-dimensional lattice.

1.3 Unit cell

A unit cell is a parallelogram (2D) or parallelepiped (3D) built from the translation vectors. Repetition of its translation generates the full lattice. A cell is *primitive* when its lattice points lie at its corners only, so that its effective count is one. A *non-primitive* cell has points at other positions inside or on its faces, so its effective count is greater than one (Fig. 1.2).

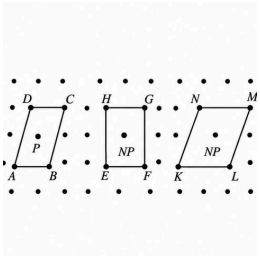


Fig. 1.2 (book): primitive (P) and non-primitive (NP) unit cells of a two-dimensional lattice.

The Wigner–Seitz cell is constructed by drawing the bisectors of the lines from a lattice point to its neighbours; the smallest enclosed volume is primitive (Fig. 1.3).

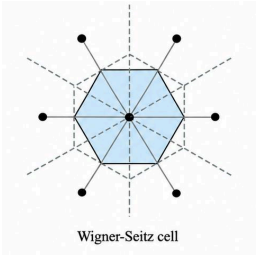


Fig. 1.3 (book): construction of the Wigner–Seitz primitive cell.

The conventional unit cell is chosen for convenience; it is the primitive cell of least volume, unless a non-primitive cell shows higher symmetry.

1.4 Basis and crystal structure

A basis is an atom or group of atoms placed identically at every lattice point. The combination is the crystal structure:

Space lattice + Basis → Crystal structure

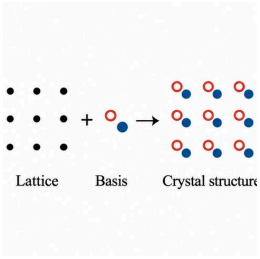


Fig. 1.4 (book): generation of a crystal structure from a lattice and a basis.

A lattice is mathematical; a crystal structure is physical. Fourteen lattices suffice because the bases vary.

> **RESULT 2: EFFECTIVE LATTICE POINTS**
>
> A lattice point at a corner is shared by eight cells (weight 1/8); at a face centre by two (1/2); a point inside belongs wholly to the cell (weight 1). Eq. (1.5) gives the count, checked in §1.7.

Summary

- Translations (Eqs. 1.1 and 1.2) define the lattice.
- Primitive cells have one effective lattice point; non-primitive cells have more.
- Crystal structure = lattice + basis.

Section 2: symmetry operations and point groups (1.5–1.6)

1.5–1.6 Symmetry operations, point groups and space groups (structured notes)

Source: book-sources/bs-1-crystal-structure.html, Puri and Babbar pp. 6–9. Equations 1.3 and 1.4 are cited by their book numbers. Figures are redrawn.

Before you start

- Lattice translation: $\mathbf{T} = n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c}$ (Eq. 1.1).
- Rotation by θ about an axis; reflection in a plane; inversion $\mathbf{r} \to -\mathbf{r}$.
- Plus Two recall: full turn 2π (box below).

PLUS TWO RECALL: FULL TURN AND INTEGRAL MULTIPLES

Recall. A full turn is 2π. A rotation that returns a figure to itself after n steps satisfies $n\theta = 2\pi$.

Re-derive.

1. Let θ be the smallest rotation that maps the lattice to itself.
2. Applying it n times gives a total turn $n\theta$, which must equal 2π (one full turn).
3. Therefore $\theta = 2\pi/n$, and n is a positive integer.

Saved. $n\theta = 2\pi$ (Eq. 1.3).

1.5 Symmetry operations

The four basic operations are:

1. Translation $\mathbf{r} \to \mathbf{r} + \mathbf{T}$, defined by Eq. (1.1). Applies to lattices only.
2. Rotation by θ about a point (2D) or axis (3D).
3. Reflection in a plane m (line in 2D).
4. Inversion $\mathbf{r} \to -\mathbf{r}$ about a centre $\bar{1}$ (3D only).

Rotation: since the lattice repeats after a full turn, the rotation angle must divide 2π into an integer number of parts. The multiplicity n satisfies

$$n\theta = 2\pi \quad \text{or} \quad \theta = \frac{2\pi}{n} \tag{1.3}$$

Compatibility with translations restricts n to 1, 2, 3, 4 and 6 (Fig. 1.5).

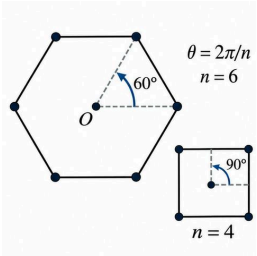


Fig. 1.5 (book): six-fold rotation about the point O in two dimensions (redrawn with a square case).

1.5.1 Proof that a five-fold axis is excluded

Consider points A, B, C, D on a row, with $\mathbf{T}$ the lattice translation between neighbours (Fig. 1.6). Rotation by $\theta = 2\pi/n$ about B (clockwise) and about C (anticlockwise) gives B' and C'. These must be lattice points, so $B'C'$ is an integral multiple of BC:

$$\begin{aligned} B'C' &= m\,(BC) \\ 2T\cos\theta + T &= mT \\ \cos\theta &= \frac{m-1}{2} \end{aligned} \tag{1.4}$$

with m an integer. The inequality $|\cos\theta| \le 1$ restricts $m \in \{3, 2, 1, 0, -1\}$, giving $\theta \in \{0°, 60°, 90°, 120°, 180°\}$ and therefore $n \in \{1, 6, 4, 3, 2\}$. The five-fold rotation is excluded.

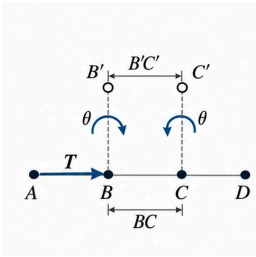


Fig. 1.6 (book): geometry used to prove that only 1, 2, 3, 4 and 6-fold rotation axes are permissible.

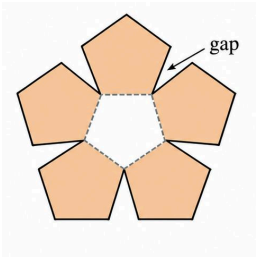


Fig. 1.7 (book): demonstration that a five-fold rotation axis cannot exist in a lattice.

1.5.2 Reflections and inversion

A reflection plane divides the lattice into mirror-image halves (Fig. 1.8).

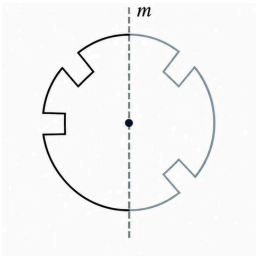


Fig. 1.8 (book): reflection symmetry of a notched wheel about a line.

Combining each of the five allowed rotations with or without a mirror gives the ten two-dimensional point groups **1, 1*m*, 2, 2*mm*, 3, 3*m*, 4, 4*mm*, 6, 6*mm*** (Fig. 1.9, book). The three-dimensional case adds the inversion $\bar{1}$ and its combinations, giving 32 point groups.

RESULT 2: COUNTS

Two dimensions: 10 point groups, 5 Bravais lattices, 17 space groups. Three dimensions: 32 point groups, 14 Bravais lattices, 230 space groups.

1.6 Point groups and space groups

A point group is the set of point operations that leave one point of the crystal fixed. The point symmetry of the crystal is that of the lattice together with the basis. A unit cell may have point symmetry at more than one location. The full set of symmetry elements, including translations, is the *space group*.

Summary

- Allowed rotation multiplicities: $n = 1, 2, 3, 4, 6$ (Eqs. 1.3 and 1.4).
- Ten 2D point groups; 32 3D point groups; 17 and 230 space groups.

Section 3: Bravais lattices, directions, planes and spacing (1.7–1.9)

1.7 Types of lattices: Bravais lattices and crystal systems (structured notes)

Source: book-sources/bs-1-crystal-structure.html, Puri and Babbar pp. 10–15. Equation 1.5 is cited by its book number. Tables 1.1 and 1.2 are reproduced from the book.

Before you start

- Point groups (Section 1.6) constrain which lattices are possible.
- Lattice parameters: a, b, c, α, β, γ.
- Sharing of lattice points between cells (recall below).

PLUS TWO RECALL: SHARING A QUANTITY AMONG CELLS

Recall. A quantity held by k identical objects is counted as $1/k$ for each.

Re-derive.

1. In a stack of cubes, a corner point touches 8 cubes; its weight per cube is $1/8$.
2. A face point touches 2 cubes; its weight per cube is $1/2$.
3. A point inside a cube belongs to that cube alone; its weight is 1.

Saved. Weights $1/8$, $1/2$ and 1 in Eq. (1.5).

1.7.1 Two dimensions

Ten point groups give five Bravais lattices in four crystal systems (Table 1.1): oblique; rectangular primitive and rectangular centred; square; hexagonal.

S. No.	Crystal system	Characteristic point group symmetry	Bravais lattice	Conventional unit cell	Unit cell characteristics
1	Oblique	1, 2	Oblique	Parallelogram	a ≠ b, γ ≠ 90°
2	Rectangular	1m, 2mm	1. Rectangular primitive 2. Rectangular centred	Rectangle	a ≠ b, γ = 90°
3	Square	4, 4mm	Square	Square	a = b, γ = 90°
4	Hexagonal	3, 3m, 6, 6mm	Hexagonal	60° Rhombus	a = b, γ = 120°

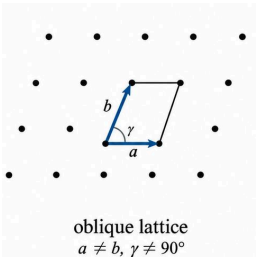


Fig. 1.10 (book): oblique two-dimensional Bravais lattice.

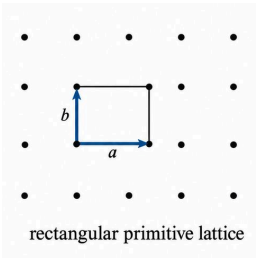


Fig. 1.10 (book): rectangular primitive two-dimensional Bravais lattice.

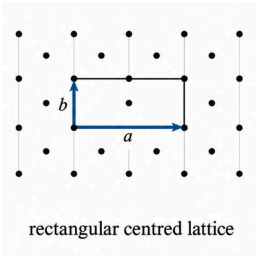


Fig. 1.10 (book): rectangular centred two-dimensional Bravais lattice.

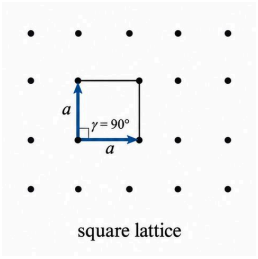


Fig. 1.10 (book): square two-dimensional Bravais lattice.

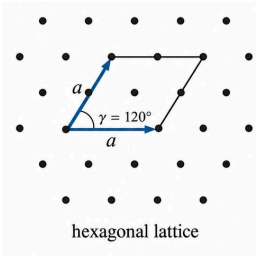


Fig. 1.10 (book): hexagonal two-dimensional Bravais lattice.

1.7.2 Three dimensions

Thirty-two point groups give fourteen Bravais lattices in seven crystal systems (Table 1.2). The symbols are P (simple), I (body-centred), F (face-centred), C (end-centred) and R (rhombohedral).

S. No.	Crystal system	Lattice parameters	Bravais lattice	Common abbreviation	Lattice symbol	Examples
1	Cubic	a = b = c; α = β = γ = 90°	Simple Body-centred Face-centred	sc bcc fcc	P I F	Cu, Ag, Fe, Na, NaCl, CsCl
2	Tetragonal	a = b ≠ c; α = β = γ = 90°	Simple Body-centred	st bct	P I	β-Sn, TiO_2

3	Orthorhombic	a ≠ b ≠ c; α = β = γ = 90°	Simple Body-centred End-centred Face-centred		so bco eco fco	P I C F	Ga, Fe_3C (cementite)
4	Rhombohedral or Trigonal	a = b = c; α = β = γ ≠ 90°	Simple		–	P (or R)	As, Sb, Bi
5	Hexagonal	a = b ≠ c; α = β = 90°, γ = 120°	Simple		–	P	Mg, Zn, Cd, NiAs
6	Monoclinic	a ≠ b ≠ c; α = γ = 90° ≠ β	Simple End-centred		– –	P C	$CaSO_4 \cdot 2H_2O$ (gypsum)
7	Triclinic	a ≠ b ≠ c; α ≠ β ≠ γ ≠ 90°	Simple		–	P	$K_2Cr_2O_7$

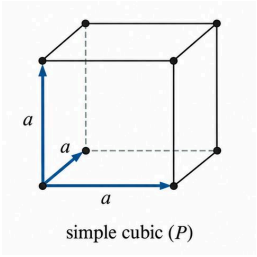


Fig. 1.11 (book): simple cubic (P) Bravais lattice.

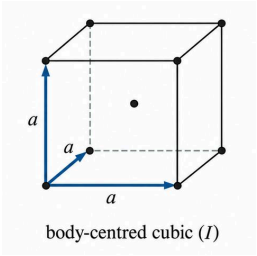


Fig. 1.11 (book): body-centred cubic (I) Bravais lattice.

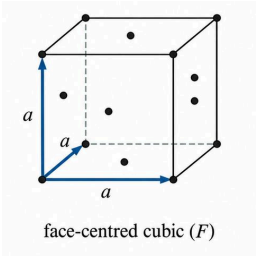


Fig. 1.11 (book): face-centred cubic (F) Bravais lattice.

1.7.3 Effective number of lattice points

$$N = N_i + \frac{N_f}{2} + \frac{N_c}{8} \qquad (1.5)$$

Results: sc $N = 1$; bcc $N = 2$; fcc $N = 4$ (with $N_c = 8$ for each, $N_i = 1$ for bcc, $N_f = 6$ for fcc).

1.7.4 Why only fourteen

The orthorhombic system has four lattices, the cubic three and the tetragonal two. The extra cells are not new arrangements: two face-centred tetragonal lattices placed side by side give a body-centred tetragonal lattice (Fig. 1.12), which is already listed.

> **RESULT: FOURTEEN BRAVAIS LATTICES**
>
> Five in two dimensions, fourteen in three. Every crystal structure is one of these lattices with a basis.

Summary

- 5 two-dimensional and 14 three-dimensional Bravais lattices.
- Eq. (1.5) gives the effective lattice points per cell.

1.8–1.9 Lattice directions, planes, Miller indices and interplanar spacing (structured notes)

Source: book-sources/bs-1-crystal-structure.html, Puri and Babbar pp. 16–20. Equations 1.6 to 1.10 are cited by their book numbers. Figures are redrawn.

Before you start

- Lattice directions and planes; the dot product.
- Plane intercept form and normal vector (box below).

1.8.1 Lattice directions

A direction is indexed by $[uvw]$: take a point on the line through the origin, read its coordinates in units of the axes, and reduce them to the smallest integers. A negative component carries a bar. Families: $\langle 100 \rangle$ (six cube edges), $\langle 110 \rangle$ (twelve face diagonals), $\langle 111 \rangle$ (eight body diagonals). Example (Fig. 1.14): AB with projections -1, 1, 0 is $[\bar{1}\,1\,0]$.

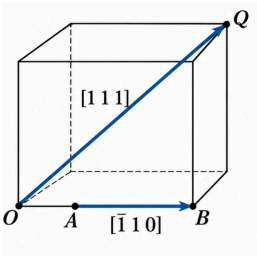
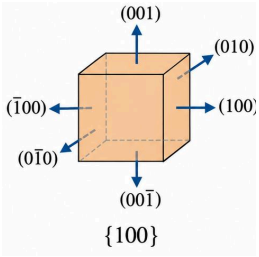


Fig. 1.13 and 1.14 (book): indices of directions in a cube (AB is [1̄10], OQ is [111]).



Family {100} of cube faces (book §1.8): six planes, six members.

1.8.2 Angle between directions

$$\cos\theta = \frac{hh' + kk' + ll'}{(h^2 + k^2 + l^2)^{1/2}\,(h'^2 + k'^2 + l'^2)^{1/2}} \tag{1.6}$$

PLUS TWO RECALL: DOT PRODUCT

Recall. $\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}| \cos\theta = \sum u_i v_i$.

Re-derive.

1. In a cubic lattice the axes are orthogonal and of equal length, so the components of $[hkl]$ in the Cartesian frame are (h, k, l) times the axis length.
2. The sum of products of components is $hh' + kk' + ll'$; each magnitude is $(h^2 + k^2 + l^2)^{1/2}$ in the same units.
3. Dividing gives Eq. (1.6).

Saved. Eq. (1.6).

1.8.3 Miller indices of planes

Steps: (i) find the intercepts in units of the axes; (ii) take reciprocals; (iii) clear fractions and write the result in round brackets, (hkl). A parallel plane has the same indices. A plane through the origin is treated by shifting the origin. Families use braces: $\{100\}$ (six planes), $\{110\}$ (six), $\{111\}$ (eight).

Worked case (Fig. 1.15): intercepts 1, 2, 1; reciprocals 1, $\frac{1}{2}$, 1; cleared 2, 1, 2; the plane is (212).

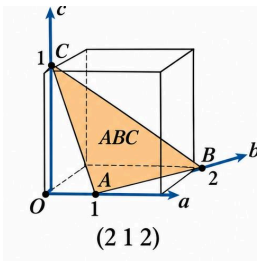


Fig. 1.15 (book): Miller indices of plane ABC, intercepts 1, 2, 1, giving (212).

1.9 Interplanar spacing

For planes (hkl) with unit normal $\hat{\mathbf{n}}$, the spacing d satisfies

$$\begin{aligned} d &= OP \\ &= \frac{a}{h}\cos\alpha \\ &= \frac{b}{k}\cos\beta \\ &= \frac{c}{l}\cos\gamma \end{aligned} \tag{1.7}$$

$$d = \frac{\hat{\mathbf{n}}\cdot\mathbf{a}}{h} = \frac{\hat{\mathbf{n}}\cdot\mathbf{b}}{k} = \frac{\hat{\mathbf{n}}\cdot\mathbf{c}}{l} \tag{1.8}$$

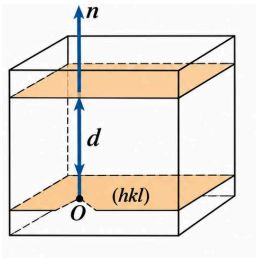


Fig. 1.17 (book): an (hkl) plane at a distance d from a parallel plane through the origin.

PLUS TWO RECALL: NORMAL AND DISTANCE OF A PLANE

Recall. For a plane $px + qy + rz = s$, the normal is (p, q, r) and the perpendicular distance from the origin is $|s|/\sqrt{p^2+q^2+r^2}$.

Re-derive.

1. Write the intercept plane as $hx/a + ky/b + lz/c = 1$: $p = h/a$, $q = k/b$, $r = l/c$, $s = 1$.
2. Normal $\propto (h/a, k/b, l/c)$.
3. $d = 1/\sqrt{(h/a)^2 + (k/b)^2 + (l/c)^2}$.

Saved. Eq. (1.9). Only the plane equation is partly recalled.

$$d = \frac{1}{\left(\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}\right)^{1/2}} \tag{1.9}$$

Valid for orthogonal lattices. For a non-orthogonal lattice, find $\hat{\mathbf{n}}$ by another method and use Eq. (1.8).

$$d = \frac{a}{\sqrt{h^2+k^2+l^2}} \tag{1.10}$$

For a cubic lattice; the direction $[hkl]$ is then normal to (hkl).

RESULT: CUBIC SPACING

$$d_{hkl} = \frac{a}{\sqrt{h^2+k^2+l^2}}$$

Example: (312) planes in an orthorhombic lattice with the intercepts of Example 1.3 (book).

Summary

- Eq. (1.6): angle between directions. Eq. (1.7)–(1.8): spacing from the normal. Eq. (1.9): orthogonal lattices. Eq. (1.10): cubic lattices.

Section 4: simple crystal structures (1.10–1.13)

1.10–1.13 Simple crystal structures (structured notes)

Source: book-sources/bs-1-crystal-structure.html, Puri and Babbar pp. 20–28 (including Examples 1.1 and 1.5). The book gives these structures without equation numbers; the displays here are untagged.

Before you start

- Coordination number and packing fraction (definitions in §1.10).
- Volume of a sphere: $\frac{4}{3}\pi r^3$ (box below).

> **PLUS TWO RECALL: VOLUME OF A SPHERE**
>
> **Recall.** A sphere of radius r has volume $\frac{4}{3}\pi r^3$.
>
> **Re-derive.**
>
> 1. Packing fraction $f = \frac{N \cdot \frac{4}{3}\pi r^3}{V_{\text{cell}}}$, with N atoms per cell.
> 2. For sc: $N = 1$, $V_{\text{cell}} = a^3$, $a = 2r$, so $f = \frac{4}{3}\pi r^3 / 8r^3 = \pi/6$.
>
> **Saved.** $f = \pi/6 \approx 0.52$ for sc.

1.10.1 Close-packed structures

Coordination number 12; six neighbours in the plane, three above and three below. Two stackings:

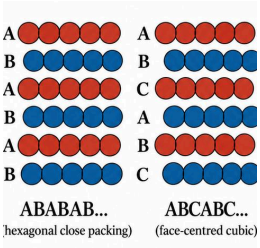


Fig. 1.18 (book): layered arrangement of close-packed structures (ABAB and ABC stacking).

hcp ($\cdots ABABAB \cdots$). Conventional cell is hexagonal (Fig. 1.19). Effective atoms per cell: $12 \times \frac{1}{6} + 2 \times \frac{1}{2} + 3 = 6$. Ideal $c = 1.633\,a$. Primitive lattice: simple hexagonal with a two-atom basis at $(0, 0, 0)$ and $(\frac{2}{3}, \frac{1}{3}, \frac{1}{2})$; primitive volume is one-third of the hexagonal cell. Packing fraction $f = \frac{2 \cdot \frac{4}{3}\pi r^3}{a^2 c \sin 60^\circ} = 0.74$.

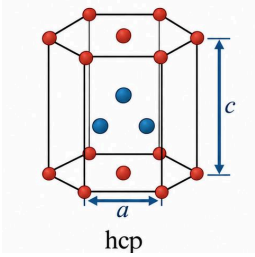


Fig. 1.19 (book): conventional and primitive cells of the hexagonal close-packed structure.

fcc ($\cdots ABCABC \cdots$). Conventional cell is a cube (Fig. 1.20), effective atoms $8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$. Touching along the face diagonal, $4r = \sqrt{2}\,a$. $f = \frac{4 \cdot \frac{4}{3}\pi r^3}{a^3} = \frac{\pi}{3\sqrt{2}} \approx 0.74$.

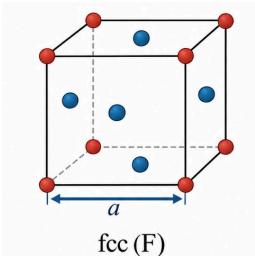


Fig. 1.20 (book): conventional unit cell of the fcc structure.

1.10.2 Loose-packed structures

bcc. Effective atoms $8 \times \frac{1}{8} + 1 = 2$; coordination 8; $4r = \sqrt{3}\,a$; $f = \frac{2 \cdot \frac{4}{3}\pi r^3}{a^3} = \frac{\pi\sqrt{3}}{8} \approx 0.68$. Examples: Na, K, Mo, W.

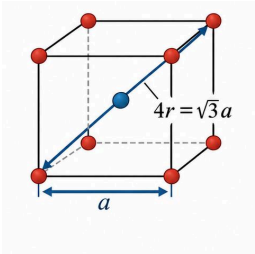


Fig. 1.21 (book): conventional unit cell of the bcc structure.

sc. Atoms at corners; $a = 2r$; coordination 6; $f = \pi/6 \approx 0.52$. Polonium at room temperature.

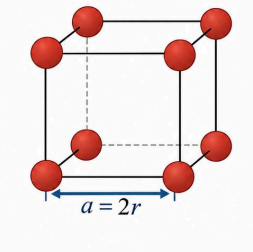


Fig. 1.22 (book): unit cell of the sc structure.

1.11 Diamond cubic structure

Space lattice fcc; basis two carbon atoms, at $(0,0,0)$ and at one quarter of the body diagonal. The second atom occupies an alternate tetrahedral void. Packing efficiency 34 %; coordination 4; nearest-neighbour distance $\sqrt{3}\,a/4$. Equivalent description: two interpenetrating fcc sublattices with origins $(0,0,0)$ and $(\frac{1}{4},\frac{1}{4},\frac{1}{4})$ (Fig. 1.24). Examples: Si, Ge, SiC, GaAs, grey tin.

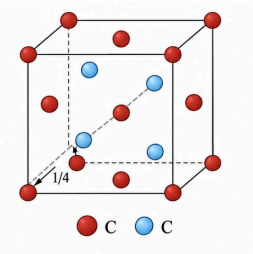


Fig. 1.23 and 1.24 (book): unit cell of the diamond cubic structure.

1.12 Zinc blende (ZnS)

As diamond cubic, but the two fcc sublattices hold different elements (Zn and S).

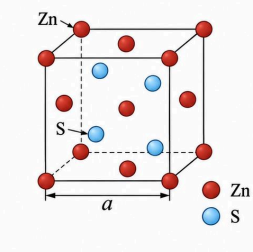


Fig. 1.23 (book), zinc blende: the diamond cubic cell with two kinds of atom.

1.13 Sodium chloride

Each Na^+ has six Cl^- octahedral neighbours. Unit cell fcc with Cl^- on the fcc sites and Na^+ in the octahedral voids. As a lattice and basis: fcc lattice, basis Na^+ at $(0,0,0)$ and Cl^- at $(a/2,0,0)$. Positions as listed in the book; four molecules per cell; $a = 2(r_+ + r_-)$.

$$\mathrm{Na^+}:\ (0,0,0);\ (\tfrac{1}{2},\tfrac{1}{2},0);\ (\tfrac{1}{2},0,\tfrac{1}{2});\ (0,\tfrac{1}{2},\tfrac{1}{2})$$

$$\mathrm{Cl^-}:\ (\tfrac{1}{2},\tfrac{1}{2},\tfrac{1}{2});\ (0,0,\tfrac{1}{2});\ (0,\tfrac{1}{2},0);\ (\tfrac{1}{2},0,0)$$

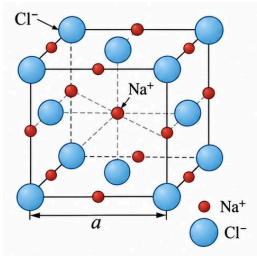


Fig. 1.25 (book): unit cell of the sodium chloride structure.

Example 1.1 and Example 1.5 (book)

Example 1.1: $a = 2r$ (sc), $a = 4r/\sqrt{3}$ (bcc), $a = 2\sqrt{2}\,r$ (fcc). Example 1.5: NaCl with $r_{\mathrm{Na}} = 0.98$ Å, $r_{\mathrm{Cl}} = 1.81$ Å gives $a = 5.58$ Å, $f = 0.663$ and density 2234 kg m^{-3}.

$$f = \frac{16\pi}{3}\cdot\frac{0.98^3 + 1.81^3}{5.58^3} = 0.663$$

> **RESULT: PACKING FRACTIONS**
>
> hcp and fcc 0.74; bcc 0.68; sc 0.52; diamond 0.34.

Summary

- Close packing: coordination 12, $f = 0.74$.
- Diamond and zinc blende: coordination 4, $f = 0.34$.
- NaCl: fcc lattice, two-ion basis, four molecules per cell.

Section 5: X-ray diffraction, reciprocal lattice and Brillouin zones (2.1–2.2, 2.4–2.7)

2.1–2.2 X-ray diffraction: Bragg's law and the Laue equations (structured notes)

Source: book-sources/bs-2-xray-reciprocal.html, Puri and Babbar pp. 34–39. Equations 2.1 to 2.5 are cited by their book numbers. Section 2.3 is outside the module. Figures are redrawn.

Before you start

- Constructive interference: path difference $n\lambda$ (box below).
- Interplanar spacing d_{hkl} and the normal to a plane (Section 1.9).

> **PLUS TWO RECALL: INTERFERENCE CONDITION**
>
> **Recall.** Constructive interference occurs for path difference $n\lambda$.
>
> **Re-derive.**
>
> 1. Rays reflect from adjacent planes of spacing d at angle θ.
> 2. Path difference = (down) + (up) = $2d\sin\theta$.
> 3. Set $2d\sin\theta = n\lambda$.
>
> **Saved.** Eq. (2.1).

2.1 Introduction

X-rays are electromagnetic waves, but ordinary gratings cannot diffract them, because their wavelength is too short. Von Laue (1912) proposed a single crystal as a three-dimensional grating. Friedrich and Knipping confirmed it with zinc blende. The Laue pattern reflects the symmetry of the crystal. X-ray diffraction determines crystal structures and the wavelength of X-rays.

2.2 X-ray diffraction

Electrons in the crystal are set oscillating by the incident beam and become secondary sources at the same frequency. The secondary waves interfere, giving maxima and minima in certain directions. Figure 2.1 shows a one-dimensional row of atoms with spacing a, where reinforcement occurs in zeroth, first and second order directions only.

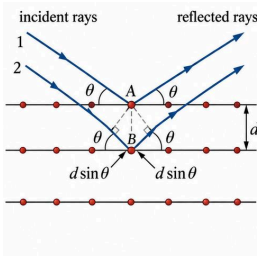


Fig. 2.1 and 2.2 (book): reinforcement of scattered waves and Bragg reflection from atomic planes.

2.2.1 Bragg's treatment

Diffraction from atoms is converted into reflection from parallel planes of atoms. For a set of planes with spacing d and glancing angle θ, the path difference is $2d\sin\theta$, and

$$2d\sin\theta = n\lambda \tag{2.1}$$

n is the order. $\sin\theta \le 1$ requires $\lambda \le 2d$.

2.2.2 Laue's treatment

Two identical scatterers A and B separated by $\mathbf{r}$ with incident and scattered unit vectors $\hat{\mathbf{n}}_1$, $\hat{\mathbf{n}}_2$ (angle 2θ between them). Path difference $= \mathbf{r}\cdot\mathbf{N}$, $\mathbf{N} = \hat{\mathbf{n}}_1 - \hat{\mathbf{n}}_2$, $|\mathbf{N}| = 2\sin\theta$ (Figs. 2.3, 2.4).

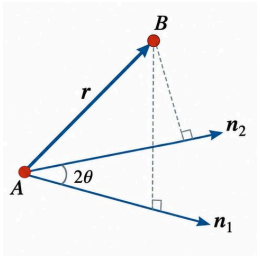


Fig. 2.3 and 2.4 (book): scattering from two identical centres; the scattering normal N.

$$\phi = \frac{2\pi}{\lambda}(\mathbf{r}\cdot\mathbf{N}) \tag{2.2}$$

Requiring constructive interference along $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ together:

$$\begin{aligned}\frac{2\pi}{\lambda}(\mathbf{a}\cdot\mathbf{N}) &= 2\pi n h' \\ \frac{2\pi}{\lambda}(\mathbf{b}\cdot\mathbf{N}) &= 2\pi n k' \\ \frac{2\pi}{\lambda}(\mathbf{c}\cdot\mathbf{N}) &= 2\pi n l'\end{aligned} \tag{2.3}$$

With $\mathbf{a}\cdot\mathbf{N} = 2a\sin\theta\cos\alpha$ and so on:

$$\begin{aligned} 2a\sin\theta\cos\alpha &= h'\lambda = nh\lambda \\ 2b\sin\theta\cos\beta &= k'\lambda = nk\lambda \\ 2c\sin\theta\cos\gamma &= l'\lambda = nl\lambda \end{aligned} \tag{2.4}$$

together with the direction-cosine identity (orthogonal axes)

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1 \tag{2.5}$$

The direction cosines of $\mathbf{N}$ are proportional to h/a, k/b, l/c, so $\mathbf{N}$ is the normal to (hkl). Using Eq. (1.7) in Eq. (2.4) gives $2d\sin\theta = n\lambda$, i.e. Bragg's law (Eq. 2.1). Laue's equations contain Bragg's law.

Convention: n is absorbed by writing $(nh\,nk\,nl)$ as the first-order reflection, so structure work uses $n = 1$ with all planes (real or imaginary) considered.

> **RESULT: EQUIVALENCE OF THE TWO TREATMENTS**
>
> The Laue conditions (2.4) with (2.5) imply $2d_{hkl}\sin\theta = n\lambda$ (Eq. 2.1).

Summary

- Eq. (2.1): Bragg's law. Eq. (2.2): phase. Eq. (2.3): three simultaneous conditions. Eq. (2.4): Laue equations. Eq. (2.5): identity.
- Diffraction requires $\lambda \leq 2d$.

2.4–2.6 Reciprocal lattice and Bragg's law in reciprocal space (structured notes)

Source: book-sources/bs-2-xray-reciprocal.html, Puri and Babbar pp. 46–54. Equations 2.10 to 2.30 are cited by their book numbers. Figures are redrawn.

Before you start

- Normals to planes; interplanar spacing (Section 1.9).
- Scalar and vector products; scalar triple product $V = \mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$.

> **PLUS TWO RECALL: DOT PRODUCT AND PERPENDICULARITY**
>
> **Recall.** A vector is perpendicular to another when their dot product is zero; $\mathbf{u}\cdot\mathbf{u} = |\mathbf{u}|^2$.
>
> **Re-derive.**
>
> 1. $\mathbf{b}\times\mathbf{c}$ is perpendicular to $\mathbf{b}$ and to $\mathbf{c}$.
> 2. Hence $\mathbf{a}^* = \mathbf{b}\times\mathbf{c}/V$ satisfies $\mathbf{a}^*\cdot\mathbf{b} = \mathbf{a}^*\cdot\mathbf{c} = 0$ (Eq. 2.15).
> 3. $\mathbf{a}^*\cdot\mathbf{a} = \mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})/V = 1$ (Eq. 2.16).
>
> **Saved.** Eqs. (2.15) and (2.16).

2.4.1 Reciprocal lattice vectors

$$\mathbf{a}_{hkl} = \frac{1}{d_{hkl}}\,\hat{\mathbf{n}} \tag{2.10}$$

The reciprocal basis of the direct basis $\mathbf{a}, \mathbf{b}, \mathbf{c}$ follows from Eqs. (2.11) and (2.12), $\mathbf{a}_{100} = (\mathbf{b} \times \mathbf{c})/V = \hat{\mathbf{n}}/d_{100}$, applied to each face:

$$\begin{aligned} \mathbf{a}^* &= \frac{\mathbf{b} \times \mathbf{c}}{V} \\ \mathbf{b}^* &= \frac{\mathbf{c} \times \mathbf{a}}{V} \\ \mathbf{c}^* &= \frac{\mathbf{a} \times \mathbf{b}}{V} \end{aligned} \tag{2.13}$$

Each reciprocal vector is normal to two direct vectors (Eq. 2.14), with the orthogonality relations (2.15) and normalisation (2.16). The book notes that $\mathbf{a}^*$ and $\mathbf{a}$ need not be parallel (monoclinic case; Fig. 2.9): then $|\mathbf{a}^*| = 1/(a \cos\theta)$.

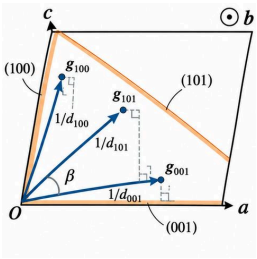


Fig. 2.8 and 2.9 (book): monoclinic cell and its two-dimensional reciprocal lattice.

In the 2π convention (Eq. 2.17, $\mathbf{a}^* \cdot \mathbf{a} = \mathbf{b}^* \cdot \mathbf{b} = \mathbf{c}^* \cdot \mathbf{c} = 2\pi$):

$$\begin{aligned} \mathbf{a}^* &= 2\pi \frac{\mathbf{b} \times \mathbf{c}}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}, \quad \mathbf{b}^* \\ &= 2\pi \frac{\mathbf{c} \times \mathbf{a}}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}, \quad \mathbf{c}^* \\ &= 2\pi \frac{\mathbf{a} \times \mathbf{b}}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})} \end{aligned} \tag{2.18}$$

Dimensions: direct $[\mathbf{length}]$, reciprocal $[\mathbf{length}]^{-1}$. Cell volumes are inversely related. Real space versus k-space.

2.4.2–2.4.4 Reciprocal of the cubic lattices

Simple cubic: $\mathbf{a}^* = \frac{2\pi}{a}\hat{\mathbf{i}}$ etc. (Eq. 2.21), so the reciprocal of sc is sc with edge $2\pi/a$.

$$\begin{aligned} \mathbf{a}^* &= \frac{2\pi}{a}\hat{\mathbf{i}} \\ \mathbf{b}^* &= \frac{2\pi}{a}\hat{\mathbf{j}} \\ \mathbf{c}^* &= \frac{2\pi}{a}\hat{\mathbf{k}} \end{aligned} \tag{2.21}$$

bcc, primitive vectors (Eq. 2.22), $V = a^3/2$; reciprocal (Eq. 2.23) is fcc with primitive vectors $\frac{2\pi}{a}(\hat{\mathbf{j}} + \hat{\mathbf{k}})$ and cyclic permutations.

$$\begin{aligned} \mathbf{a} &= \frac{a}{2}(\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}), \quad \mathbf{b} \\ &= \frac{a}{2}(-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}), \quad \mathbf{c} \\ &= \frac{a}{2}(\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}) \end{aligned} \tag{2.22}$$

$$\begin{aligned} \mathbf{a}^* &= \frac{2\pi}{a}(\hat{\mathbf{j}} + \hat{\mathbf{k}}), \quad \mathbf{b}^* \\ &= \frac{2\pi}{a}(\hat{\mathbf{i}} + \hat{\mathbf{k}}), \quad \mathbf{c}^* \\ &= \frac{2\pi}{a}(\hat{\mathbf{i}} + \hat{\mathbf{j}}) \end{aligned} \tag{2.23}$$

fcc, primitive vectors (Eq. 2.24), $V = a^3/4$; reciprocal (Eq. 2.25) is bcc with edge $4\pi/a$.

$$\begin{aligned} \mathbf{a}' &= \frac{a}{2}(\hat{\mathbf{i}} + \hat{\mathbf{j}}), \quad \mathbf{b}' \\ &= \frac{a}{2}(\hat{\mathbf{j}} + \hat{\mathbf{k}}), \quad \mathbf{c}' \\ &= \frac{a}{2}(\hat{\mathbf{k}} + \hat{\mathbf{i}}) \end{aligned} \tag{2.24}$$

$$\begin{aligned} \mathbf{a}^* &= \frac{2\pi}{a}(\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}), \quad \mathbf{b}^* \\ &= \frac{2\pi}{a}(-\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}), \quad \mathbf{c}^* \\ &= \frac{2\pi}{a}(\hat{\mathbf{i}} - \hat{\mathbf{j}} + \hat{\mathbf{k}}) \end{aligned} \tag{2.25}$$

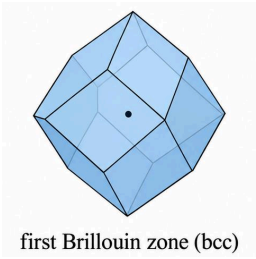


Fig. 2.15 (book): first Brillouin zone of a bcc lattice.

2.4.3 Lattice points and the reciprocal lattice

Direct points $\mathbf{T} = m\mathbf{a} + n\mathbf{b} + p\mathbf{c}$ (Eq. 2.19). Reciprocal points $\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*$ (Eq. 2.20). Since $\mathbf{G} \cdot \mathbf{T} = 2\pi(hm + kn + lp)$, $\exp(i\mathbf{G} \cdot \mathbf{T}) = 1$. Each reciprocal point (hkl) corresponds to the planes (hkl).

$$\mathbf{T} = m\mathbf{a} + n\mathbf{b} + p\mathbf{c} \tag{2.19}$$

$$\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^* \tag{2.20}$$

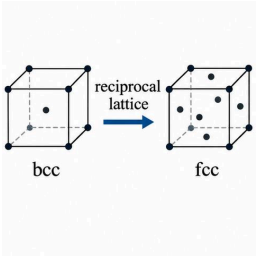


Fig. 2.10 and 2.11 (book): bcc and fcc primitive vectors, whose reciprocals are each other.

2.5 Properties

1. Each reciprocal point is one set of parallel direct planes.
2. Its distance from the origin is inversely proportional to that spacing.
3. Reciprocal cell volume is inversely proportional to the direct cell volume.
4. The Wigner–Seitz cell of the reciprocal lattice is the Brillouin zone.
5. The direct lattice is the reciprocal of its reciprocal; sc is self-reciprocal; bcc and fcc are reciprocal to each other.

2.6 Bragg's law in reciprocal space

Ewald construction: AO of length $1/\lambda$ to origin O; sphere of radius AO about A.

$$|\mathbf{OB}| = \frac{n}{d_{hkl}} \tag{2.26}$$

$$\mathbf{OB} = \frac{2\sin\theta}{\lambda} \tag{2.27}$$

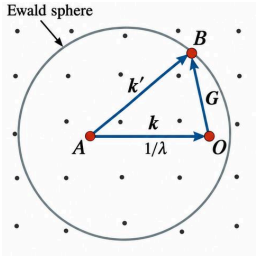


Fig. 2.12 (book): Ewald construction in the reciprocal lattice.

Equating: $2d_{hkl}\sin\theta = n\lambda$. Magnifying by 2π gives $\mathbf{k}' = \mathbf{k} + \mathbf{G}$, $k'^2 = k^2$, and

$$2\mathbf{k}\cdot\mathbf{G} + G^2 = 0 \tag{2.28}$$

$$\begin{aligned} \mathbf{k}' &= \mathbf{k} + \mathbf{G} \\ k'^2 &= k^2 \\ \mathbf{k}' - \mathbf{k} &= \Delta\mathbf{k} = \mathbf{G} \end{aligned} \tag{2.29}$$

Eq. (2.30) states the last of these: the change in $\mathbf{k}$ is the reciprocal vector $\mathbf{G}$.

> **RESULT**
>
> Eq. (2.28) is the vector form of Bragg's law and is used for the Brillouin zones.

Summary

- Eqs. (2.13), (2.18): reciprocal basis. Eqs. (2.21), (2.23), (2.25): cubic reciprocals.
- Eq. (2.28): Bragg's law in vector form.

2.7 Brillouin zones (structured notes)

Source: book-sources/bs-2-xray-reciprocal.html, Puri and Babbar pp. 55–58 (to the end of 2.7.2; section 2.8 is outside the module). Equations 2.31 to 2.34 are cited by their book numbers. Figures are redrawn.

Before you start

- Bragg's law in vector form, Eq. (2.28).
- Reciprocal lattice vectors of sc, bcc and fcc (Eqs. 2.21, 2.23, 2.25).

> **PLUS TWO RECALL: PERPENDICULAR BISECTOR**
>
> **Recall.** Points equidistant from two points $\mathbf{0}$ and $\mathbf{G}$ satisfy $\mathbf{k}\cdot\mathbf{G} = G^2/2$.
>
> **Re-derive.**
>
> 1. $|\mathbf{k}|^2 = |\mathbf{k}-\mathbf{G}|^2$.
> 2. $2\mathbf{k}\cdot\mathbf{G} = G^2$.
>
> **Saved.** The normal bisector of $\mathbf{G}$.

2.7 Definition

The Brillouin zone is the locus of wave vectors $\mathbf{k}$ that are Bragg reflected by the reciprocal lattice (Ewald construction).

2.7.1 Square lattice

With $\mathbf{a}^* = (2\pi/a)\hat{\mathbf{i}}$, $\mathbf{b}^* = (2\pi/a)\hat{\mathbf{j}}$, $\mathbf{G} = (2\pi/a)(h\hat{\mathbf{i}} + k\hat{\mathbf{j}})$ and $\mathbf{k} = k_x\hat{\mathbf{i}} + k_y\hat{\mathbf{j}}$, Eq. (2.28) gives

$$hk_x + kk_y = -\frac{\pi}{a}(h^2 + k^2) \qquad (2.31)$$

$(h, k) = (\pm 1, 0)$ gives $k_x = \pm\pi/a$; $(0, \pm 1)$ gives $k_y = \pm\pi/a$. These bound the first zone, a square of side $2\pi/a$ (Fig. 2.14). $(\pm 1, \pm 1)$ gives $\pm k_x \pm k_y = 2\pi/a$, bounding the second zone.

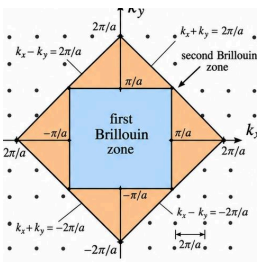


Fig. 2.14 (book): Brillouin zones of a square lattice in its reciprocal lattice.

Zone boundaries are reflecting planes of order given by the zone number. A $\mathbf{k}$ off the boundaries is not reflected.

2.7.2 Cubic lattices in three dimensions

$$hk_x + kk_y + lk_z = -\frac{\pi}{a}(h^2 + k^2 + l^2) \qquad (2.32)$$

First zone: cube of side $2\pi/a$. Method of normal bisectors: first zone bounded by bisectors of the shortest $\mathbf{G}$; second zone by the next shortest, and so on.

2.7.3 bcc zone

$\mathbf{G} = \frac{2\pi}{a}[(h+l)\hat{\mathbf{i}} + (h+k)\hat{\mathbf{j}} + (k+l)\hat{\mathbf{k}}]$ (Eq. 2.33). Shortest $\mathbf{G}$: twelve vectors $(2\pi/a)(\pm\hat{\mathbf{i}} \pm \hat{\mathbf{j}})$ and cyclic. Zone: rhombic dodecahedron (Fig. 2.15).

$$\mathbf{G} = \frac{2\pi}{a}[(h+l)\hat{\mathbf{i}} + (h+k)\hat{\mathbf{j}} + (k+l)\hat{\mathbf{k}}] \qquad (2.33)$$

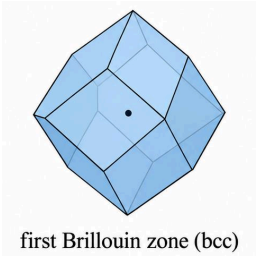


Fig. 2.15 (book): first Brillouin zone of a bcc lattice.

2.7.4 fcc zone

$\mathbf{G} = \frac{2\pi}{a}[(h-k+l)\hat{\mathbf{i}} + (h+k-l)\hat{\mathbf{j}} + (-h+k+l)\hat{\mathbf{k}}$ (Eq. 2.34). Eight shortest $\mathbf{G}$: $(2\pi/a)(\pm\hat{\mathbf{i}} \pm \hat{\mathbf{j}} \pm \hat{\mathbf{k}})$. Corners of the octahedron truncated by $(2\pi/a)(\pm 2\hat{\mathbf{i}})$ and cyclic. Zone: truncated octahedron (Fig. 2.16), also a primitive cell of bcc.

$$\mathbf{G} = \frac{2\pi}{a}[(h-k+l)\hat{\mathbf{i}} + (h+k-l)\hat{\mathbf{j}} + (-h+k+l)\hat{\mathbf{k}}] \quad (2.3$$

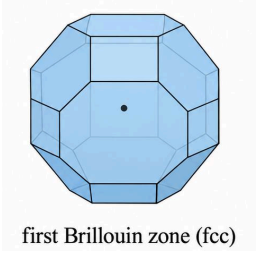


Fig. 2.16 (book): first Brillouin zone of an fcc lattice.

> **RESULT**
>
> sc: cube, $2\pi/a$. bcc: rhombic dodecahedron. fcc: truncated octahedron.

Summary

- Eqs. (2.31)–(2.34): zone conditions for square, cubic, bcc and fcc.